MILITARY SPECIFICATION

NUT RAISIN MIX (Operational Ration Component)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers nut raisin mix for use by the Department of Defense as a component of Ration, Cold Weather.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-44276 - Packaging of Ration, Cold Weather Components (in pouches).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8925

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(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Regulations Governing the Inspection and Grading of Eggs and Egg Products

(Application for copies should be addressed to the Director, Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Grades of Shelled Runner, Virginia and Spanish Type Peanuts

U.S. Standards for Grades of English Walnuts

U.S. Standards for Grades of Shelled Almonds.

U.S. Standard Grades of Filberts in the Shell.

U.S. Standards for Grades of Processed Raisins

(Application for copies should be addressed to the Chief, Fresh Products Inspection, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists Inc., 3340 Pilot Knob Road, St. Paul, MN 55121.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in a contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off odors, off flavors, and off colors.

3.2.1 Nuts, peanuts, shelled. Shelled peanuts shall be USDA certified U.S. No. 1 Runner, Virginia, or Spanish type whole peanuts of the United States Standards for Shelled Runner, Virginia and Spanish Type Peanuts. The peanuts shall be blanched and the skins removed. The blanched peanuts shall be properly dry roasted without scorching or burning. The peanuts shall be protected by the types and amounts of antioxidants approved by the FDA for peanuts except that the final product mix shall not have antioxidants exceeding approved limits. The shelled, blanched, roasted peanuts shall be from the latest season's crop.

3.2.2 Raisins. Raisins shall be Thompson seedless select size (1000 to 1200/pound) and be USDA certified U.S. Grade B or better of the U.S. Standards for Grades of Processed Raisins. Moisture content of the raisins shall be between 13 and 15 percent as tested by a Dried Fruit Association moisture meter. Raisins shall be oil coated with a 500-hour (AOM) hydrogenated vegetable oil (Durkex 500 manufactured by Durkee Foods or equivalent). Raisins shall be from the latest season's crop.

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3.2.3 Nuts, walnuts, English, shelled, pieces. Shelled walnut pieces shall be USDA certified U.S. Commercial Grade or better of the U.S. Standards for Grades of English Walnuts except that the shell count shall not exceed 5 per 25 pounds and the walnut fiber count shall not exceed 7 per 25 pounds. Walnut pieces shall be an amber color or lighter and the piece size shall be such that no more than 25 percent pass through 24/64 inch round holes, no more than 5 percent (included in the 25 percent) pass through 16/64 inch round holes and no more than 1 percent (included in the 5 percent) pass through 8/64 inch round holes. Walnut pieces shall be protected by the types and amounts of antioxidants approved by the FDA for walnuts except that the final product mix shall not have antioxidants exceeding approved limits. Walnut pieces shall be from the latest season's crop.

3.2.4 Nuts, almonds, shelled. Shelled Non Pariel or California almonds shall be USDA certified U.S. Select Sheller or better of the U.S. Standards for Grades of Shelled Almonds. The almonds shall be either 23 to 25 inclusive, 24 to 26 inclusive, or 26 to 28 inclusive, count range per ounce. The shelled almonds shall be from the latest season's crop.

3.2.5 Nuts, filberts, shelled. Shelled filberts shall be USDA certified, from U.S. No. 1 of the U.S. Standards for Grades of Filberts in the Shell and shall be small or medium size such that their diameter is between 11 and 13 millimeters or approximately a nut count of between 30 and 40 nuts per ounce. Shelled filberts shall be from the latest season's crop.

3.2.6 Shellac, edible. Edible shellac coating shall be a high grade, clear amber shellac. The liquid coating shall contain pharmaceutical glaze, ethyl alcohol, and acetylated monoglyceride and shall have a solids content of not less than 23 percent and a specific gravity of not less than 0.83 (see 6.3).

3.2.7 Coating, corn protein. Corn protein coating shall be a natural edible coating consisting of the corn protein zein solubilized in ethyl alcohol containing suspended vegetable oil (partially hydrogenated soybean and cottonseed oils). The solution shall also contain glycerine, citric acid, BHA, and BHT. The liquid coating shall contain 37 percent solids (see 6.4).

3.2.8 Coating, glaze. Glaze coating shall be an approximate 5 to 2 ratio mixture of sugar and dry egg whites. The sugar shall be white refined, granulated cane or beet sugar, or a combination thereof. The dry egg whites shall contain no whipping aids and shall be processed and labeled in accordance with the Regulations Governing the Inspection and Grading of Eggs and Egg Products (7 CFR Part 59) as evidenced by the USDA egg products inspection shield on the label. Each lot of dry egg whites shall be sampled and analysed for salmonella by the USDA as evidenced by a USDA Egg Products Inspection Certificate. Hot water shall be used as the solvent.

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3.3 Preparation and processing.

3.3.1 Formulation of the nut raisin mix. The nut raisin mix shall be formulated as follows:

Ingredients	Percent by weight
Peanuts, whole, roasted	65.0
Raisins	15.0
Walnuts, pieces	10.0
Almonds, whole	5.0
Filberts, whole	5.0

3.3.2 Preparation of nut raisin mix. Uniformly coat peanuts and walnuts separately either by spray or dipper application while product (peanuts and walnuts) tumbles in a revolving pan or cylinder with either the edible shellac, corn protein, or glaze coating material. Peanuts and walnuts shall be further dryed such that residual solvent is fully evaporated and individual nuts are separated. Edible shellac shall be applied at between 3 and 4 percent by weight, corn protein coating at between 0.3 and 1.0 percent by weight, and glaze at between 15 and 20 percent by weight. Nuts and Raisins shall either be blended together in the formulated proportions or filled directly into pouches in the formulated proportions.

3.4 Pouch filling and sealing. Each pouch shall be filled with a minimum of 54.0 grams and a maximum of 60.0 grams of mix and sealed as specified in MIL-P-44276.

3.5 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, paper, paint, glass, or metal.
b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, moldy, or carmelized.
c. There shall be no foreign color in the product.
d. Moisture content of the total finished product (nut and raisin mixture) shall not be greater than 5.0 percent.
e. Final blended product shall be free of clumped nuts and raisins such that product is free flowing and clumps can be broken with light finger pressure.
f. Final packaged product shall contain peanuts: 62 to 67 percent, raisins: 13.0 to 17.0 percent, walnuts: 8.0 to 12.0 percent, almonds and filberts: 4.0 to 7.0 percent each.

3.5.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

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3.6 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency."

3.7 Federal Food, Drug and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined

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organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, and filling requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Results shall be reported to the nearest 0.1 gram. Any individual net weight of less than 54.0 grams or more than 60.0 grams shall be scored as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

4.5.4 Product inspection. The finished product shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects.

TABLE I. Product defects 1/ 2/

Category	Defect
Major	
101	Product not free flowing

1/ The presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, paper, paint, or metal), foreign odor or flavor (e.g. burnt, scorched, stale, sour, rancid, musty, moldy or carmelized), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

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4.5.5 Ingredient weight inspection. Twenty filled and sealed pouches shall be selected at random from each lot. The contents of the 20 pouches shall be weighed and individual ingredients separated and weighed separately. The percent of each ingredient shall be determined and reported to nearest 0.1 percent. Any result falling outside the applicable range specified in 3.5.f shall be cause for rejection of the lot and shall be a major defect.

4.5.6 Moisture content testing. The moisture content of total mix shall be determined in accordance with the Nut and Nut Products chapter, vacuum oven method of the Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC). Results shall be reported to the nearest 0.1 percent. Any result not conforming to the moisture content requirement in 3.5.d. shall be scored as a major defect. The lot size shall be expressed in units of pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.7 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44276.

5. PACKAGING

5.1 Packaging. Packaging of the raisin mix shall be in accordance with the applicable requirements of MIL-P-44276.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.2).
c. Provisions for approved preproduction samples (see 3.5.1 and 6.2).

6.2 First article inspection. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Edible shellac. The edible shellac used in product development was OPAGLOS GS-2-0400 Amber manufactured by Colorcon, Inc.

6.4 Coating, corn protein. The corn protein coating used in product development was COZEEN 303N manufactured by Zumbro, Enterprises.

6.5 Subject term (key word) listing.

Operational rations
Ration, Cold Weather

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Custodians:

Army - GL

Review activities:

Army - MD, TS
Navy - MC, MS, SA
Air Force - 50
DP - SS

Preparing activity:

Army - GL

Project No. 8925-0137

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-N-44280(GL)	NUT RAISIN MIX (Operational Ration Component)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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